

UDK Phase 1: Data Strategy

Sep 29, 2026 · @Sohan · Baseline: GitHub main @ 96ea038 (UDK Phase 0: Baseline Specification)

1. Phase 0 → Phase 1 problem definition

Phase 1 answers one question: what data does the next system need, and where does it come from. No model is trained, tuned or replaced here; every number below about the baseline is taken from the Phase 0 spec (main @ 96ea038), and every dataset fact carries a source link.

The Phase 0 failures that data must let us study, in Phase 0's own IDs:

Phase 0 ID	Failure	Baseline evidence (Phase 0)	What blocks progress today
F2, F12	Overlapping / second-speaker speech lost	UDK_03 + UDK_09 missed 6/6 conditions; TTS overlap 1/5	One recorded overlap scenario; no real overlap test set
F3, F4	Indic speech becomes invented English; no code-switching	Whisper returns English on 95–99% of FLEURS clips; Telugu dev 4.2/h	No Indic or code-switched danger audio; recall never measured
F5–F7, F9	Right words, harmless context (quotes, stories, negation, fragments)	Quote/story control alerts 6/6; "Come on now" scores 0.999	Too few real hard negatives; none Indian
F9, F10	No speaker identity; context model v2 silences cries after attacker speech	v2 probe: "Get away!" after attacker line 0.998 → 0.000	No speaker-attributed attacker→victim data, real or synthetic
F11	Scream trigger false alarms; frightened speech missed	2.4/h sealed, 18.7/h gameplay; 28/57 real distress sounds missed; fear speech BEATs 0.013–0.022	Hard negatives thin; no Indian environments; no frightened-speech set

Phase 0 ID	Failure	Baseline evidence (Phase 0)	What blocks progress today
F13	Pocket / muffled / clipped audio	Pocket 12/14; crowd condition is filtered white noise	Device conditions are simulated only
F15	Evaluation sets tuned on; harmless-only sealed sets	Only 7 real danger media clips; sealed sets measure false alarms only	No frozen real danger test set
— (§8)	Custom UDK never evaluated	Denver placeholder live; no numbers	No personal-phrase recordings at all

The central finding of this phase, stated up front: public data can cover ASR, code-switching, conversation structure, overlap mechanics and acoustic hard negatives. It cannot cover the project's core problem — the same words in safe and unsafe contexts, spoken by attributed speakers, in Indian languages — so that data must be collected.

2. Data requirements matrix

Each row maps a Phase 0 problem to the data needed to study it. Priority follows section 21: P0 blocks Phase 2, P1 is important, P2 useful, P3 optional.

#	Phase 0 problem	Data required	Why required	Minimum characteristics	Priority
A	Multilingual ASR (F4)	Transcribed real speech, several Indian languages	Indic speech turns into invented English	Native-script + English transcripts; speaker IDs; spontaneous share	P0
B	Indian-language speech (F4)	Real spontaneous Indic speech, many districts	Recall in any Indian language never measured	≥50 speakers/language for eval; gender + region metadata	P0
C	Hindi (F4)	Hindi spontaneous + telephone speech	Largest user language	Accent spread; phone-band subset	P0

#	Phase 0 problem	Data required	Why required	Minimum characteristics	Priority
D	Telugu (F4)	Telugu spontaneous speech	Own recordings already alert 4.2/h	Andhra + Telangana speakers; English words kept	P0
E	English, Indian-accented (F2, F13)	Indian-English speech with WER references	English WER never measured	Many accents, read + spontaneous	P1
F	Hinglish (F4)	Hindi-English code-switched speech	Mixed utterances lose meaning in English-only ASR	Intra-sentence switching; Latin + Devanagari kept	P0
G	Telugu-English (F4)	Telugu-English code-switched speech	Telugu false alarms come from English words inside Telugu	Word-level language tags	P0
H	Other Indian code-switching	Bengali/Tamil/Kannada/Marathi-English	Coverage beyond Hi/Te	Eval-size sets	P2
I	Conversational speech (F9, F10)	Spontaneous dialogue with turns	Context model trained on generated sentences	Turn boundaries, timestamps	P1
J	Multi-speaker (F10)	Speaker-labelled conversations	No speaker identity anywhere	Speaker IDs per word/segment	P1
K	Overlap (F12)	Real overlapping speech with per-speaker references	Main recall gap	Overlap timestamps; per-speaker transcripts	P0
L	Far-field / pocket (F13)	Phone-in-pocket/bag recordings	Pocket is weakest condition (12/14)	Real devices, not filters	P0
M	Phone-band	Telephone speech	Phone condition simulated	8 kHz real calls	P2
N	Traffic noise (F13)	Real Indian traffic audio	Traffic false alarms 3.9/h	Horns, two-wheelers, buses	P1

#	Phase 0 problem	Data required	Why required	Minimum characteristics	Priority
O	Crowd noise	Real babble, markets, stations	Crowd condition is white noise	Real multi-talker babble	P1
P	Clipping	Real clipped phone audio	Loud condition simulated	Device AGC/clipping	P2
Q	Low-SNR speech	Speech at measured SNRs	Quiet/crowd conditions synthetic	SNR labels	P1
R	Safety utterances (F5-F9)	Spoken distress phrases, real speakers	Only 7 real danger clips	Per-UDK coverage, many speakers	P0
S	Same phrase safe vs unsafe (F5, F9)	Minimal pairs, target text fixed	The project's central problem	Pair IDs; context-only difference	P0
T	Negation	"I don't need help" vs "I need help"	Veto relies on it	Spoken, not only text	P1
U	Quotation	Quoted UDKs	Quote control alerts 6/6	Spoken; speaker marked	P0
V	Story / narration	UDKs inside stories	Same	Spoken	P0
W	Third-person discussion	"She said he was attacking her"	Same	Spoken	P1
X	Attacker → victim (F10)	Speaker-attributed sequences	v2 silenced cries after attacker lines	Speaker IDs; real voices	P0
Y	Victim → attacker	Reverse order	Symmetry for the context model	Same	P1
Z	Distress without UDK words	Frightened speech, no phrase	BEATs 0.013–0.022 on fear speech	Real or acted, many speakers	P1
AA	Scream (F11)	Real screams, varied	11/25 FSD50K screams caught	Live vs recorded marked	P1
AB	Crying	Real crying	16/25 caught	Adult crying, not only baby	P1
AC	Whimpering	Whimper / sobbing	Class in live trigger	Small but real	P2

#	Phase 0 problem	Data required	Why required	Minimum characteristics	Priority
AD	Frightened speech	Fear-prosody speech	Weak-UDK rule parked for lack of it	Indian speakers	P1
AE	Ordinary shouting	Calling out, arguing	Hard negative for scream trigger	Indian settings	P1
AF	Laughter	Laughter clips	1-3/25 calibration FPs	Varied	P1
AG	Cheering	Crowds, sports	Same	Indian events	P1
AH	Gaming/TV/movie audio	Media with screams	18.7/h in gameplay	Train only where licensed	P0
AI	Everyday Indian conversation	Long harmless audio	False-alarm rate per hour	Hours, not clips; licensed	P0
AJ	Long-duration natural audio	Full journeys	Live re-segmentation differs from eval	30-60 min continuous	P1
AK	Custom/personal UDK	Users' own phrases, safe + unsafe	Never evaluated; placeholder live	Per-user phrase, many speakers	P1

3. Stream A — multilingual ASR data

Two commercially usable corpora carry this stream: IndicVoices (natural, 22 languages) and Vaani (spontaneous, district-level). Read-speech sets are kept for evaluation or pronunciation coverage only, never as the sole source.

Read vs natural speech. FLEURS, Kathbath and Common Voice are prompted read speech. Phase 0 used FLEURS to show the English-only failure; that is a valid failure test but it says nothing about spontaneous Indian speech, disfluency or mixed-language turns. Natural sets below are marked as such.

Dataset	Languages	Size	Speakers	Speech type	Transcripts	License (class)	Source
IndicVoices (AI4Bharat)	22 Indian	23.7K h collected, 11.2K h transcribed (latest card)	51K, 400+ districts	8% read, 76% extempore, 15% conversational	Native script	CC BY 4.0 (B, attribution)	HF card , paper
Vaani (IISc/ARTPARK, Google)	105 languages/dialects	~31,278 h audio, 2,122 h transcribed	156K, 165 districts	Spontaneous, image-prompted	Local script, ~10% transcribed	CC BY 4.0, gated (B)	HF card
Gram Vaani ASR (Interspeech 2022)	Hindi, regional variants	100 h + 5 h + 3 h labelled, 1,000 h unlabelled	NOT VERIFIED	Spontaneous telephone	Crowd transcripts, variable accuracy	Academic free; commercial needs permission (C)	OpenSLR 118
Kathbath (AI4Bharat)	12 incl. Hindi, Telugu, Tamil, Kannada, Malayalam, Bengali, Marathi	1,684 h	1,218, 203 districts	Read	Native script	CC0 packaging (A)	HF card
Lahaja (AI4Bharat)	Hindi, multi-accent	12.5 h	132, 83 districts	Read + extempore	Yes	CC BY 4.0 (B)	page , HF
Svarah (AI4Bharat)	Indian-accented English	9.6 h	117, 19 states	Read + spontaneous	Yes	LICENSE NOT VERIFIED (D)	GitHub
SPRING-INX (IIT Madras)	10 Indian (no Telugu)	~2,000 h	NOT VERIFIED	Mixed domains	Manual	Reported CC BY 4.0, NOT VERIFIED at source (D)	paper
MUCS 2021 sub-task 1	Hi, Mr, Or, Ta, Te, Gu	~600 h total incl. sub-task 2	NOT VERIFIED	Mostly read	Single script per transcript	NOT VERIFIED for SLR103 (D)	challenge

What each needs checked by inspection before Phase 2 (metadata only, no training):

- Script of English words.** Commercial Indic ASR vendors document that native-script output transliterates English words inside Indic speech ([example model card](#)). Whether IndicVoices and Vaani transcripts do the same is NOT VERIFIED; it decides whether "attack" in "mujhe koi attack kar raha hai" survives as a recoverable token.
- Device and channel.** Vaani and IndicVoices are phone-collected per their project pages; exact device mix and sample rates are NOT VERIFIED.

3. **Telugu depth.** Vaani lists about 184 transcribed Telugu hours across districts (card table); IndicVoices Telugu hours to be counted from its manifest.
4. **Gram Vaani** is the only large real telephone set; its commercial use requires written permission from Gram Vaani.

4. Stream B — speaker / conversation / context data

Public corpora can teach conversation structure (who spoke when, turn-taking, overlap) but none pairs speaker turns with danger scenarios. They are useful for the speaker-attribution and diarization side of F10; the attacker/victim content must be collected (section 15).

Dataset	Language	Size	Speakers	Speaker labels / turns	Overlap	Environment	License (class)	Source
DISPLACE 2023/2024 (IISc, NITK)	7-9 Indian languages + Indian English, code-mixed	~32 h (2023); 38 h annotated (2024)	3-5 per session, 30-60 min sessions	Speaker turns, language turns, transcripts	Natural overlaps	Single-channel far-field, social conversation	Challenge use only; no sharing with non-participants (E)	2023 summary , 2024 , 2026 card
AMI Meeting Corpus	English, mostly non-native	100 h	Meetings of 3-4	Word-level, per-speaker headset + far-field	Yes	3 meeting rooms	CC BY 4.0 per official site (B)	AMI, license
ICSI Meeting Corpus	English	NOT VERIFIED	Multi-party	Yes	Yes	Meeting room	CC BY 4.0 per AMI/ICSI site (B)	AMI/ICSI page
NOTSOFAR-1 (Microsoft)	English	315 meetings of ~6 min; ~170 h single-channel device audio; 1,000 h simulated train	35 unique, 4-8 per meeting	Word-level, per speaker	Yes, real	30 office rooms, commercial devices	LICENSE NOT VERIFIED (D)	paper , repo
MELD	English (TV series Friends)	~13,000 utterances, 1,433 dialogues	Multi-party	Speaker names per utterance; emotion incl. fear	Scripted	TV audio	Repository license NOT VERIFIED; audio is copyrighted TV content (F for training)	paper , repo

Discrepancy to flag. OpenSLR's AMI mirror still lists a modified CC BY-NC-SA 2.0 licence ([OpenSLR 16](#)); the official AMI site states CC BY 4.0. Download from the official site and keep its licence file.

What Stream B cannot supply. None of these contains a threat followed by a plea, a friend's check-in followed by a plea, or narration of such a scene. AMI and NOTSOFAR are office meetings; DISPLACE is social conversation and access-restricted; MELD is scripted comedy. Phase 0's own attempt to mine this (ConVAWG, KCDD, DKTC) also failed (status doc §17).

5. Stream C — safety intent / hard negatives

No public dataset holds spoken same-text / different-context pairs for personal danger, in any language. The closest public resources give either emergency speech without harmless twins, or negation text without speech. Stream C is therefore mostly a collection stream (section 15), seeded by synthetic text (section 16).

Resource	What it has	Language	Real vs synthetic	Hard negatives?	License (class)	Use	Source
Korean Emergency Audio and Speech (AI Hub)	~3,500 h, 16 emergency situations incl. call-for-help, harassment, violent crime	Korean	Reported as recorded scenarios; acted vs real NOT VERIFIED	Not as minimal pairs	Access restricted, NOT VERIFIED (E)	Reference design only	described in
CARES corpus	Simulated emergency dialogues, phrases, spontaneous speech; 40 actors aged 23–91	English (Canada)	Acted	No	NOT VERIFIED (D)	Design reference	survey
HIS distress-call subset	20 distress phrases in a smart home	French	Acted	No	NOT VERIFIED (D)	Design reference	survey
CANNOT	77,376 sentence pairs, ~half negated, half paraphrased	English text	Derived from existing datasets + generated paraphrases	Negation only	Dataset licence NOT VERIFIED (companion model is CC BY-SA 4.0) (D)	Text-only negation eval	HF card
Project's own intent datasets v1/v2	6,587 and 18,503 rows	English + invented English	Generated	Negation pairs; v2 context rows skewed	Own	Regression only; v2 context rows not reusable as-is (Phase 0 §11)	Phase 0 §10–11
Mined movie/chat candidates (intent_data/mined/)	Unlabelled	Mixed	Real text	Unknown	Source licences unknown (D)	Label for eval only if sources are cleared	Status doc §17

Required pair structure. Every Stream C item is stored as a pair with the target words fixed and only context varying:

pair_id	target_text	context (before)	context speaker	target speaker	label
P0001	Help me	Can you help me carry these boxes?	A	A	SAFE
P0001	Help me	Don't move. Nobody is coming to help you.	B (attacker)	A (victim)	DANGER
P0002	Mujhe help karo	Is bag ko uthane mein mujhe help karo.	A	A	SAFE
P0002	Mujhe help karo	Mujhe yahan se jaane nahi de raha hai.	A	A	DANGER
P0003	Don't touch me	(story) "...and she shouted, don't touch me"	A (narrator)	A	SAFE
P0003	Don't touch me	Come here, nobody can hear you.	B (attacker)	A (victim)	DANGER

The target utterance must be recorded separately from its context so the same audio can be re-used across contexts for controlled tests, and also recorded in-scene so natural prosody is captured. Both versions are labelled with which method produced them.

6. Stream D — acoustic distress data

Positives (screams, cries) exist publicly but mostly under non-commercial licences; the licence-clean route is FSD50K filtered to CC0/CC-BY clips. Hard negatives matter more than positives here, because the live trigger's false alarms come from media screams, gameplay and crowds (Phase 0 §9).

Dataset	Positives	Hard negatives	Size	Real / acted	Indian?	License (class)	Source
FSD50K	Screaming, crying/sobbing, whimper, shout, yell	Laughter, cheering, crowd, siren, horn, music	51,197 clips, 108.3 h, 200 classes	Real (Freesound uploads)	No	Dataset CC BY; per-clip CC0 / CC-BY / CC-BY-NC / Sampling+ (B after filtering out NC)	card , paper

Dataset	Positives	Hard negatives	Size	Real / acted	Indian?	License (class)	Source
AudioSet	527 classes incl. Screaming, Crying, Battle cry	Video game music, cheering, laughter	~2M YouTube clips	Real	Some	Labels public; audio not distributed, only features; re-download raises usage-rights issues (F for training)	FSD50K paper on AudioSet
IUEC (IIIT-Delhi)	Screams, cries	Indoor, outdoor (metro, road horn, market), crowds, cheering, TV (cricket, Indian songs, comedy)	~20 h context set + 250 h of 17 volunteers' daily audio	Mixed: phones, movies, internet	Yes (Delhi)	EULA required; terms NOT VERIFIED (E)	project page
Nonspeech7k	Screaming, crying	Laughing, coughing, breathing, sneezing, yawning	7 classes, ~7k clips	Real (Freesound-sourced)	No	CC BY-NC-SA 4.0 (C)	paper
BERSt	Shouted and fear-prompted speech	Normal speech, other emotions	4,526 clips, ~3.75 h, 98 actors, 19 phone positions	Acted, smartphones in homes	No (accents varied)	CC BY 4.0 (B)	paper , card
ASVP-ESD	Fear (scream, panic), cry, pain (groan)	Laugh, neutral, excitement	12,625 samples	Mixed sources	Unknown	NOT VERIFIED (D)	described in
VIVAE	Intense non-speech vocalisations incl. fear	Pleasure, surprise	1,085 samples, 11 speakers	Acted	No	NOT VERIFIED (D)	described in
IITKGP-SESC / SEHSC	Fear, anger (acted)	Neutral, happy, sarcasm	~12,000 utterances each, 10 radio artists	Acted	Telugu / Hindi	Distribution terms NOT VERIFIED (D)	SEHSC , SESC
Vaani Noise Event	Baby/child crying and yelling (as background)	Horns, engines, sirens, trains, music, prayer, TV, appliances	90,637 segments, ~154.6 h; 10 h speaker-disjoint eval held back by authors	Real, speech with background events	Yes, many districts	CC BY 4.0, gated (B)	card

Live vs recorded screams. Phase 0 lists this as unresolved (§19 item 4). No dataset found labels a scream as live versus played through a speaker. This needs collected pairs: the same scream performed live and replayed through a phone/TV speaker, recorded on the same device (section 15).

Frightened speech without screaming. Only acted sets exist (BERSt fear prompts, IITKGP fear). They can support evaluation of the Z/AD rows but are not evidence of real fear; collection with a clear acted-label is required.

7. Indian language data

Hindi and Telugu are well covered for general ASR; neither has any public danger speech. Coverage per target language, from the Stream A sources:

Language	Natural / spontaneous	Read	Telephone	Code-switched	Gap
Hindi	IndicVoices, Vaani (largest transcribed language)	Kathbath, Lahaja	Gram Vaani (C)	MUCS Hi-En, HiACC	Danger speech; Hinglish conversation
Telugu	IndicVoices, Vaani (~184 h transcribed per card table)	Kathbath	None found	IndicVoices subset only (see §8)	Everything safety-related; Te-En labelled set
English (Indian)	Svarah (spontaneous part)	Svarah	None found	—	English WER references in noise
Tamil, Kannada, Malayalam, Bengali, Marathi	IndicVoices, Vaani	Kathbath	—	MUCS Bn-En (Bengali)	Eval-only sets suffice for P2

Sources: [IndicVoices](#), [Vaani](#), [Kathbath](#), [Lahaja](#), [Svarah](#), [Gram Vaani](#).

Phase 0 recorded that Indic STT code (te, kn, hi, ta) exists but is disabled and its 95–100% accuracy claims have no traceable evaluation data (§6). The sets above give that code a verifiable evaluation later; choosing a model is Phase 2's decision.

8. Hinglish / code-switching data

Natural, conversational Hinglish exists publicly only in small or restricted sets; Telugu–English has no dedicated public speech corpus. The largest open Hindi–English set is lecture speech about computer science, not conversation.

Dataset	Pair(s)	Natural vs prompted	Word-level language labels	Context	Size / speakers	License (class)	Source
MUCS 2021 sub-task 2	Hi-En, Bn-En	Natural lecture speech (spoken tutorials); switching driven by technical terms	Two scripts in transcript (script = language proxy)	Monologue	Hi-En 89.86 h train + 5.18 h test; Bn-En 46.11 h + 7.02 h	CC BY-SA 4.0 (B, share-alike)	OpenSLR 104
DISPLACE	Hi, Kn, Bn, MI, Te, Ta + Indian English	Natural social conversation, 3-5 speakers	Language-turn annotations	Full conversations	~32-38 h	Challenge-only (E)	paper
IndicVoices code-switched subset	Te-En, Hi-En, Mr-En, Bn-En	Extempore / conversational	None; selected by Code-Mixing Index	Short utterances	~2.1-2.5 h eval each, 85-124 speakers (CoSTA selection)	CC BY 4.0 (B)	CoSTA paper
HiACC	Hi-En, adults + children	Spontaneous answers + story reading	NOT VERIFIED	Single speaker	5.24 h, 3,318 adult + 1,858 child segments	NOT VERIFIED (D)	paper
HingCoS (IIT Guwahati)	Hi-En	Prompted read sentences	NOT VERIFIED	None	~25 h, 101 speakers, 8 kHz	NOT VERIFIED (D)	paper

Transcript representation (section 6 of the brief). Target form for every code-switched item: the full utterance, each token in the script of its language, plus a word-level tag. Example:

Token	help	me	mujhe	koi	attack	kar	raha	hai
Lang	en	en	hi	hi	en	hi	hi	hi

A romanised copy is kept alongside so text models see one script. What must be checked per source before use: whether English words are kept in Latin or transliterated (MUCS: two scripts per the task paper, [arXiv](#)); IndicVoices and Vaani: NOT VERIFIED).

Telugu-English. The CoSTA authors state no existing Te-En speech-transcription-translation set existed and built a ~2.3 h evaluation slice from IndicVoices ([source](#)). That slice is the only public Te-En starting point found; a labelled Te-En set must be collected.

9. Overlapping speech data

Real overlap with per-speaker references exists in English meetings (AMI, NOTSOFAR) and Indian social conversation (DISPLACE, restricted). None has the Phase 0 failure case: a quieter victim phrase under a louder second speaker, on a phone in a pocket. That case is collected, and simulated mixes are built only from licence-clean sources.

Source	Overlap type	Per-speaker reference	Far-field / device	Noise	License	Role
AMI	Natural meeting overlap	Headset per speaker	Room arrays	Low	CC BY 4.0 (B)	Train + eval of overlap handling
NOTSOFAR-1	Natural office overlap	Close-talk headsets (train/dev)	Commercial devices, 30 rooms	Real room noise	NOT VERIFIED (D)	Eval only until licence checked
DISPLACE	Natural, multilingual	Speaker turns	Single-channel far-field	Real	Challenge-only (E)	Research eval only, if access granted
LibriMix / SparseLibriMix	Simulated 2-3 speaker, variable overlap	Clean sources	None	WHAM! noise	LibriSpeech speech CC BY 4.0; WHAM! noise CC BY-NC 4.0 (C with noise)	Scripts reusable; replace WHAM! noise with licence-clean noise

Sources: [AMI licence](#), [NOTSOFAR-1](#), [DISPLACE](#), [LibriMix](#), [WHAM! licence note](#).

Required overlap conditions for the D_overlap set (collected): male over female and female over male; target 0, -6, -12 dB relative to the other voice; target phrase fully inside vs partly outside the other speaker's turn; pocket, bag and hand; quiet room, street, crowd.

10. Real-world Indian audio

Two public sources carry real Indian environments with licence terms we can read: Vaani (speech recorded in 165 districts with real background) and the Vaani Noise Event set (timestamped horns, traffic, sirens, music, TV, children). IIIT-Delhi's IUEC adds Delhi street, metro and media contexts under an EULA. Everything else Indian and real — buses, trains, markets at scale, college, restaurants — was not found publicly and is collected.

Environment	Public source	Can train?	Can evaluate?
Street / traffic	Vaani Noise Event (vehicle_traffic); IUEC outdoor	Vaani: yes (B). IUEC: per EULA	Yes
Home	Vaani (appliance, TV, children categories); IUEC indoor	Vaani: yes	Yes
Market	IUEC (market sample)	Per EULA	Per EULA
Metro / train	IUEC (metro); Vaani (train events)	Partly	Yes
Bus, restaurant, college, office	None found	—	Collect
Media (TV, songs, cricket)	IUEC multimedia; Vaani singing_music	Vaani: yes	Yes
Long continuous days	IUEC volunteer set: 250 h, 17 volunteers, no distress	Per EULA	Yes (false alarms/h)

Sources: [Vaani Noise Event](#), [IUEC](#).

YouTube. Phase 0's dev and sealed YouTube sets stay evaluation-only (false alarms per hour). They are not training data: licence terms of individual videos are unknown, and the sealed set is frozen (section 18).

11. Emergency / crisis audio

The only real emergency-call audio available is the project's current US 911 set, and its licence is unclear; no Indian 112/helpline audio was found publicly. Emergency audio is therefore evaluation-only and treated as acoustic/context evidence, not as representative of Indian speech.

Candidate	Country / language	Real vs acted	Content	Size	Metadata	License / privacy	Use
911 Recordings (Kaggle, louisteitelbaum)	US, English	Real calls, collected from public sources by the late editor of Dispatch Monthly	Critical or unusual calls, caller + dispatcher	700+ calls (Phase 0: 707; 355 dev / 352 sealed)	Date, state, deaths, potential death, false alarm	Licence NOT VERIFIED; real people in crisis (D, privacy review)	Eval only; sealed half stays sealed
911 Recordings: first 6 seconds	US, English	Real	Caller's first 6 s after starting to explain	Subset	Same	Same (D)	Eval of caller-only speech (removes dispatcher words that inflate Phase 0's 194/220)
Korean Emergency Audio and Speech	Korea, Korean	NOT VERIFIED	16 emergency situations	~3,500 h	Scenario labels	Access restricted (E)	Not used
CARES	Canada, English	Acted	Emergency dialogues, older adults	40 actors	Age	NOT VERIFIED (D)	Design reference
India 112 / women's helpline audio	India	—	—	—	—	Not public	Not available; any partnership needs legal review

Sources: [911 Recordings](#), [first 6 seconds](#), [Korean dataset as described](#), [CARES](#).

The caller-only subset directly addresses Phase 0's note that dispatcher speech inflates the 911 result.

12. Dataset comparison tables

No overall ranking is given. Table A is the required evaluation table; Table B scores each serious candidate on independent dimensions (H/M/L/?). Licence classes: A commercial, B commercial with conditions, C non-commercial, D unclear, E restricted, F unsuitable.

Table A — evaluation table

Dataset	Language	Hours	Speakers	Natural speech	Code-switching	Speaker labels	Overlap	Context	Safety relevance	License	Train?	Eval?	Main limitation
IndicVoices	22 Indian	11.2K transcribed	51K	Yes (91%)	Some	Speaker IDs	No	Short	None	B	Yes	Yes	English-word script unverified
Vaani	105	2,122 transcribed	156K	Yes	Some	Speaker metadata	No	Image talk	None	B	Yes	Yes	Only ~10% transcribed
Vaani Noise Event	Many Indian	154.6	?	Yes	Some	?	No	No	Hard-neg noise	B	Yes	Yes	Background events, no distress
Gram Vaani	Hindi	105 labelled	?	Yes, phone	No	?	No	No	None	C	No (without permission)	Yes	Noisy crowd transcripts
Kathbath	12 Indian	1,684	1,218	No (read)	No	Yes	No	No	None	A	Yes	Yes	Read speech only
Lahaja	Hindi	12.5	132	Partly	No	Yes	No	No	None	B	Eval pref.	Yes	Small
Svarah	Indian English	9.6	117	Partly	No	Yes	No	No	None	D	No	Yes	Licence unverified
MUCS Hi-En / Bn-En	Hi-En, Bn-En	148	?	Lecture	Yes	?	No	Monologue	None	B (SA)	Yes	Yes	Technical-lecture domain
DISPLACE	7-9 Indian	~38	3-5/session	Yes	Yes	Yes	Yes	Full conv.	None	E	No	If granted	Challenge-only terms
AMI	English	100	Meetings	Yes	No	Yes	Yes	Full conv.	None	B	Yes	Yes	Office, non-Indian

Dataset	Language	Hours	Speakers	Natural speech	Code-switching	Speaker labels	Overlap	Context	Safety relevance	License	Train?	Eval?	Main limitation
NOTSO FAR-1	English	~170 device h	35	Yes	No	Yes	Yes	Full conv.	None	D	No	Yes	Licence unverified
FSD50K (CC0/BY subset)	—	<108	—	Real sounds	—	—	—	—	Screams + hard negs	B	Yes	Yes	NC clips must be removed
BERSt	English	3.75	98	Acted	No	Yes	No	No	Shouted/fear speech	B	Yes	Yes	Acted, nonsense phrases
IUEC	Indian contexts	~270	17 volunteers + clips	Real + media	—	Volunteer IDs	—	Daily life	Screams/cries + contexts	E	Per EULA	Per EULA	EULA, part movie-sourced
Nonspeech7k	—	?	?	Real sounds	—	—	—	—	Screams, cries	C	No	Yes	Non-commercial
911 Recordings	US English	? (707 calls)	Callers + dispatchers	Real	No	No	Yes	Call	Real emergencies	D	No	Yes	Licence, privacy, not Indian
MELD	English	? (13K utt.)	TV cast	Scripted	No	Yes	Some	Dialogue	Fear label only	F	No	Research only	Copyrighted TV audio

Table B — suitability dimensions (H high, M medium, L low, ? unknown)

Dataset	Language cov.	Natural	Speaker div.	Code-switch	Context	Overlap	Noise realism	Safety rel.	Annotation	Transcript	Licence	Size	Domain sim.
IndicVoices	H	H	H	M	L	L	M	L	M	H	H	H	M
Vaani	H	H	H	M	L	L	H	L	M	M	H	H	M
Vaani Noise Event	M	H	?	L	L	L	H	M	H	M	H	M	H
Gram Vaani	L	H	M	L	L	L	H	L	L	L	L	M	M
Kathbath	H	L	H	L	L	L	L	L	H	H	H	H	L
MUCS Hi-En	L	M	?	H	L	L	L	L	M	M	M	M	L
DISPLACE	M	H	M	H	H	H	M	L	H	M	L	L	M
AMI	L	H	M	L	H	H	M	L	H	H	H	M	L

Dataset	Language cov.	Natural	Speaker div.	Code-switch	Context	Overlap	Noise realism	Safety rel.	Annotation	Transcript	Licence	Size	Domain sim.
FSD50K subset	—	H	?	—	L	L	M	M	M	—	M	M	L
BERSt	L	L	H	L	L	L	M	M	H	H	H	L	M
IUEC	L	M	L	—	M	?	H	H	M	—	L	M	H
911 Recordings	L	H	H	L	H	M	H	H	L	L	L	L	L

13. Dataset → Phase 0 failure mapping

Every public source addresses a mechanism, never the full failure; the right-hand column is the reason section 15 exists.

Phase 0 failure	Dataset	What it provides	What remains missing
F4 Indic speech → invented English	IndicVoices, Vaani	Real spontaneous Hindi/Telugu + transcripts, many districts	Danger phrases; English-word script to verify
F4 recall in Indian languages unmeasured	None public	—	Spoken Hindi/Telugu/code-switched UDKs: collect
F4 code-switching	MUCS Hi-En; IndicVoices CS slice; DISPLACE	Mixed-language transcripts	Conversational Hinglish at scale; any labelled Te-En; safety content
F3 hallucination on noise / non-English	Vaani Noise Event, Kathbath, IndicVoices	Audio where English-only ASR produces text that is not there	Nothing further needed for measurement
F2/F12 overlap recall	AMI, NOTSOFAR, DISPLACE	Real overlap with per-speaker references	Quieter victim under louder speaker; pocket; Indian; safety speech
F9/F10 context without speakers	AMI, DISPLACE (turn structure)	Speaker turns, timestamps	Attacker→victim and friend→victim scenes: collect

Phase 0 failure	Dataset	What it provides	What remains missing
F5/F7 quotes, stories, near-miss wording	CANNOT (text negation)	Negation pairs	Spoken quotes/stories with UDK words; Indian languages
F11 scream false alarms	FSD50K subset, Vaani Noise Event, IUEC	Screams + laughter, cheering, horns, TV, music	Live vs replayed screams; gameplay in Indian homes
F11 frightened speech missed	BERSt, IITKGP (acted)	Acted fear/shouted speech on phones	Real or realistic fear from Indian speakers
F13 pocket / muffling	BERSt (19 phone positions incl. obstructed)	Real device positions	Pocket/bag specifically; Indian speech; overlap
F15 real danger recall unmeasured	911 caller-only subset	Real emergency speech	Indian; licence; privacy
§8 custom UDK never evaluated	None public	—	Personal phrases, safe + unsafe: collect

14. Public data gaps

Seven categories were searched for and not found publicly under usable terms. Each is marked COLLECT OUR OWN rather than forcing a poor substitute.

Category	Found publicly?	Closest public item	Decision
Indian safety conversations	No	Korean emergency set (restricted, Korean)	COLLECT OUR OWN
Indian code-switched safety speech	No	MUCS Hi-En (lectures)	COLLECT OUR OWN
Same-text / different-context pairs (spoken)	No	CANNOT (English text negation)	COLLECT OUR OWN + synthesize text seeds

Category	Found publicly?	Closest public item	Decision
Attacker→victim / friend→victim interactions	No	MELD speaker turns (TV, unsuitable)	COLLECT OUR OWN
Indian distressed speech (fear, pleading)	No (acted Hindi/Telugu emotion only)	IITKGP-SEHSC/SESC	COLLECT OUR OWN
Indian overlapping distressed speech	No	DISPLACE (social, restricted)	COLLECT OUR OWN
Custom UDK speech	No	—	COLLECT OUR OWN
Live vs replayed screams	No	—	COLLECT OUR OWN
Real pocket/bag recordings of Indian speech	No	BERSt (English, obstructed phone)	COLLECT OUR OWN

What public data does solve: general Indic and Indian-English ASR evaluation and training, Hindi-English code-switched transcription (lecture domain), meeting-style overlap mechanics, Indian background noise, and English acoustic hard negatives.

15. Proposed custom data collection

Collection is designed here, not started. It records scripted and semi-scripted scenes with consenting adult speakers, always as SAFE/DANGER twins of the same target phrase, across languages, speaker set-ups and environments.

Speakers.

- Adults 18–65 only; target $\geq 60\%$ women (the product's users), all genders represented.
- Age bands 18–25, 26–40, 41–65, each $\geq 20\%$ of speakers.
- Regions: Telangana, Andhra Pradesh, Hindi belt (UP/Bihar/MP), Delhi NCR, Maharashtra, Karnataka, Tamil Nadu; accents recorded as metadata.
- Minimum 200 speakers for the first release; speaker IDs pseudonymous, assigned before recording, never reused across splits (section 18).

Languages and code-switching patterns. English (Indian), Hindi, Telugu, Hinglish, Telugu-English as P0; Tamil-, Kannada-English as P2. Patterns: English UDK inside Indic sentence ("please meri help karo"), Indic UDK inside English sentence, inter-sentence switch, fully monolingual.

Devices and placements. At least 6 Android + 2 iOS phone models across price bands; placements: hand, trouser pocket, jacket pocket, handbag, table 1 m / 3 m, active phone call.

Target phrases. The 20 standard UDKs in English plus natural Hindi and Telugu equivalents written by native speakers (not machine translation); per-user custom phrases (AK) with each speaker choosing 1–3 of their own.

Protocol per target phrase. Record the target three ways: (1) isolated, (2) inside a SAFE scene, (3) inside a DANGER scene. Scenes are written by scenario writers, performed by 1–3 actors, and each utterance is labelled with speaker ID, role (victim, attacker, friend, narrator, bystander, media) and timestamps.

Hard negatives inside the same sessions. Laughter, ordinary shouting, arguing, cheering (cricket), games and films played on a phone or TV nearby, children playing, calling out to someone across a street.

Live vs replayed. A subset of screams, cries and UDKs performed live, then the same recording replayed through a phone speaker and a TV at the same distance.

Annotation. Two independent annotators per item; disagreement adjudicated. Fields: full code-switched transcript, word-level language tags, romanised copy, speaker turns, overlap spans, SNR estimate, placement, device, environment, label and pair_id.

Scenario matrix (collection blueprint). Every cell of Context × Speaker configuration is recorded in each P0 language, then spread across environments so each environment gets every context at least once.

Context	Label	Single speaker	Two speakers	Multi-speaker	Overlapping	Attacker → victim	Victim → attacker
Genuine distress	DANGER	Yes	Yes	Yes	Yes	Yes	Yes
Emergency (third party hurt)	DANGER	Yes	Yes	Yes	Yes	—	—
Harmless request ("help me carry")	SAFE	Yes	Yes	Yes	Yes	—	—
Quotation	SAFE	Yes	Yes	—	—	—	—

Context	Label	Single speaker	Two speakers	Multi-speaker	Overlapping	Attacker → victim	Victim → attacker
Storytelling / narration	SAFE	Yes	Yes	Yes	—	—	—
Movie/game dialogue playing nearby	SAFE	Yes	Yes	Yes	Yes	Media voice	Media voice
Joking / play-fighting	SAFE	—	Yes	Yes	Yes	Yes (playful)	Yes (playful)
Hypothetical ("what if someone...")	SAFE	Yes	Yes	—	—	—	—
News discussion	SAFE	Yes	Yes	Yes	—	—	—
Third-person report	SAFE	Yes	Yes	Yes	—	—	—
Negation ("I don't need help")	SAFE	Yes	Yes	—	—	—	—
Negation but danger ("I'm not okay, help")	DANGER	Yes	Yes	—	—	—	—
Instruction ("shout help if...")	SAFE	Yes	Yes	—	—	—	—
Argument, no threat	SAFE	—	Yes	Yes	Yes	—	—

Environments (each context ≥ 1 per environment): quiet room, home with TV, street/traffic, bus/auto, crowd/market, restaurant, college, pocket, bag, phone call, low SNR (≤ 0 dB), clipping (shouted close to mic).

First-release size target, to be confirmed after a pilot of 20 speakers: $\geq 3,000$ SAFE/DANGER pairs across P0 languages, ≥ 300 overlap scenes, ≥ 20 hours of long continuous harmless Indian audio from consenting participants.

Three safety problems, kept separate. Every collected or labelled item carries `problem_type`, and metrics are always reported per type, never pooled.

problem_type	Definition	Example	Evaluated on	Priority
A_standard_udk	One of the 20 fixed UDKs (or its native Hindi/Telugu equivalent)	"Help me"	SAFE/DANGER pairs below	P0
B_custom_udk	The speaker's own enrolled phrase	a user-chosen phrase	Custom UDK set (row AK)	P1
C_danger_no_udk	Danger stated without any UDK words	"He has a knife."	Danger-without-UDK set (row Z) + matched harmless mentions	P1

Phase 0 measured only type A; B and C have no baseline numbers (Phase 0 §8, §19).

Proposed minimum pair allocation (type A, P0). This is a proposed collection TARGET, derived from the $\geq 3,000$ -pair target above; it is not a collected result. A pair = one SAFE and one DANGER recording of the same target text by the same speaker. Context families group the scenario-matrix rows; each pair's SAFE side is drawn from the family named, its DANGER side from genuine distress or emergency.

Context family (SAFE side)	English	Hindi	Telugu	Hinglish	Telugu-English	Minimum total
Harmless request ("help me carry")	120	140	140	120	80	600
Quotation, story, third-person report	120	140	140	120	80	600
Movie/game/TV playing nearby	90	105	105	90	60	450
Negation (incl. negation-but-danger on the DANGER side)	90	105	105	90	60	450
Joking, play-fighting, argument without threat	90	105	105	90	60	450
Hypothetical, instruction, news discussion	90	105	105	90	60	450
Language minimum	600	700	700	600	400	3,000
Actually collected	0	0	0	0	0	0

Within the 3,000, speaker configuration minimums apply across the whole set, not per cell: $\geq 30\%$ of DANGER sides are two-speaker scenes with attacker $\rightarrow$ victim or friend $\rightarrow$ victim order; $\geq 10\%$ involve overlapping speech (these also count toward the 300 overlap scenes). Rationale for the split: Hindi and Telugu weigh more because they are the product's main languages and have zero measured recall (Phase 0 §6); Telugu-English is smallest because no public reference exists to calibrate it, and it is expanded after the pilot.

Types B and C, proposed minimums (P1): custom UDK ≥ 50 speakers $\times$ 2 own phrases $\times$ SAFE/DANGER; danger without UDK ≥ 300 DANGER utterances across P0 languages, each with a harmless utterance on the same topic ("the knife is in the kitchen drawer").

Provenance labels. Every safety/context item carries exactly one `provenance` value:

provenance	Meaning	Planned in this corpus?
acted_scripted	Actors read a written scene	Yes, main source of pairs
acted_semi_scripted	Actors given a situation and target phrase, words improvised	Yes, for natural prosody and turn-taking
spontaneous_safe	Unscripted real harmless speech	Yes, D_spontaneous_safe (section 18)
real_incident	Recording of an actual dangerous event	No; only with separate legal/ethics approval and consent
unknown	Provenance cannot be established (e.g. mined media)	Evaluation/reference only

Acted danger is used deliberately for controlled experiments: it lets the same target text appear in both contexts. Results on acted danger are reported as acted and are not claims about real-world danger recall.

Annotation schema: context and timestamps. Primary formulation is `context_before` $\rightarrow$ `target`, the real-time case. An optional `context_after` field stores what was said after the target. Phase 2 may use only preceding context; `context_after` exists for offline analysis and later experiments and does not imply the production system waits for future audio. This is a schema capability, not an architecture decision.

Field	Required	Notes
pair_id, problem_type, label, provenance	Yes	
target_text, target_speaker_id, target_role	Yes	role: victim, attacker, friend, narrator, bystander, media
context_before (text, speaker IDs, roles)	Yes	may be empty for isolated targets
context_after	Optional	offline use only
segment_start/end, target_start/end	Yes	ms from recording start
turn_start/end per speaker turn	Yes	ordered; turn index kept
overlap_start/end, overlapping speaker IDs	Yes where overlap	
context_start/end	Yes	span used as context_before
alert_event time	Where available	from later system runs, never from collection

Latency is defined conceptually for Phase 2, not measured now: audio event (target_start) → VAD segment closed → ASR text available → context decision → alert. Each interval is computable from the timestamps above plus system logs.

16. Synthetic data strategy

Synthetic data is allowed for text seeds, augmentation and mixes; it never replaces validation on real audio, and every synthetic item carries a `synthetic=true` flag and its generator ID. Phase 0 already showed the cost of ignoring this: v2's LLM context rows taught "you" = harmless (§11), and the "crowd" condition was filtered white noise.

Use	Permitted as	Real-data check required	Main risk
ASR error simulation	Train (text model inputs)	Compare error types to real ASR on collected audio	Errors unlike real Whisper/Parakeet output
Code-switching augmentation (text)	Train seeds; scripts for actors	Native-speaker review of every template	Unnatural switch points
Hard-negative generation (text)	Scripts for collection; train with cap	Must appear in collected audio too	Template leakage, repetition
Negation minimal pairs	Train (already in v1)	Spoken negation set in D_hard_negative	Over-reliance on "not" token
Context minimal pairs	Scripts only until balanced	Attacker/victim roles balanced per the v2 lesson	Shortcuts (you/he/she, length, names)
Noise augmentation	Train	Real noisy collected audio in eval	Filtered noise $\neq$ real environments
Overlap augmentation	Train	Real overlap scenes in D_overlap	Level/timing unlike real conversation
TTS speech	Not for positives	—	No panic prosody (Phase 0 §13)

Rules:

1. Synthetic items are template-split: no template appears in more than one split.
2. Generated text is checked for the Phase 0 shortcuts before use: pronoun balance per label, sentence length per label, named entities per label.
3. No model under evaluation, or its training outputs, generates its own test items.
4. All evaluation sets named D_* in section 18 are real audio, except where a set is explicitly marked synthetic (for regression only).

17. ASR error dataset strategy

A separate resource, D_asr_errors, pairs real audio with reference transcripts and the actual output of the frozen baseline engines, so text models can later be trained and tested on realistic STT mistakes rather than invented ones.

Construction. Run the frozen Whisper small.en and Parakeet TDT 0.6B v2 (no changes) over licensed audio with references: IndicVoices, Vaani, Kathbath, MUCS Hi-En, Svarah/Lahaja (eval only), and all collected audio. This is inference for data creation, not training.

Stored per sample.

Field	Example
audio_id, source, licence	ivoices_te_0123, IndicVoices, CC BY 4.0
reference transcript (full, mixed script)	help me mujhe koi attack kar raha hai
whisper_text, avg_logprob, temperature	help me attack
parakeet_text	help me
language, code-switch tags	hi+en; en en hi hi en hi hi hi
error types	deletion; language switching
condition	pocket, 0 dB, overlap

Error types labelled: substitution, deletion, insertion, hallucination (text with no speech), language switching (Indic rendered as English words), word-boundary error, name/entity error, negation error ("not" dropped or added).

Guardrails. Collected danger audio used for D_final_test is never run into D_asr_errors training rows; its ASR outputs exist only inside the test set. Split by speaker, as the audio it came from.

18. Train / dev / test strategy

The primary split is speaker-disjoint; scenario, template, environment and source splits are layered on top where the data allows. Splits are assigned by a deterministic hash of speaker ID (collected data) or of the source's own speaker/session ID (public data), written to a manifest before any audio is inspected.

Set	Contents	Real / synthetic	Split rule	Used for
D_train	Public train portions + collected train speakers + synthetic text	Both, flagged	Speaker-disjoint from all others	Fitting
D_dev	Collected dev speakers + public dev splits	Real	Speaker-disjoint	Threshold/model selection
D_hard_negative	Quotes, stories, jokes, media, negation, harmless requests	Real	New speakers, new wording, new contexts; never used in training	FA evaluation
D_overlap	Collected overlap scenes	Real	Speaker- and scene-disjoint	Overlap recall
D_multilingual	Hindi, Telugu, other Indic eval (IndicVoices/Vaani/Kathbath test, Lahaja, collected)	Real	Source test splits + held-out speakers	ASR + recall per language
D_code_switch	MUCS Hi-En test/blind, IndicVoices CS slice, collected Hi-En/Te-En	Real	Speaker-disjoint	CS ASR + recall
D_acoustic	FSD50K eval (licence-filtered), Vaani Noise held-out, collected live/replayed	Real	Source eval splits + held-out speakers	Scream/cry trigger
D_final_test	Stratified sample of every collected cell + the existing sealed sets	Real	Frozen before Phase 2 model selection; totals only	Final decision

Existing Phase 0 evaluation data stays sacred. prepped_data, real_recordings, dev YouTube, Telugu own recordings, chess audio, sealed YouTube and both 911 halves are never used to train. The sealed YouTube set and 911 sealed half go into D_final_test unchanged; dev sets remain regression checks (Phase 0 §22).

Hard-negative held-out design. If training has "Help me" in contexts {boxes, homework, cooking}, D_hard_negative uses new speakers, new wordings of the target ("please help", "koi help karo") and new contexts {game, film, story}.

D_spontaneous_safe — formal false-alarm evaluation set (P0). Long continuous harmless Indian audio from consenting participants (provenance spontaneous_safe), evaluated as its own set, not used as noise. It adds to, and does not replace, the Phase 0

dev and sealed YouTube sets.

Measure	Reported per
False alarms per hour (speech path, scream path, total, as Phase 0 §15)	environment, language/code-switch, device placement
Alert episodes on media playback, normal shouting, laughter, cheering, conversation	event type (from annotation)
Behaviour over long recordings	recording, in 30-min windows

Minimum required coverage (current coverage: 0 h): ≥ 20 h total; ≥ 30 participants; each recording ≥ 30 min continuous; ≥ 6 environments with ≥ 2 h each (home with TV, street/traffic, bus/auto, market/crowd, office/college, restaurant); ≥ 3 h each of Hindi, Telugu, English and code-switched speech; ≥ 3 h with media playing; annotated spans of shouting, laughter and cheering present in ≥ 3 environments. Split speaker-disjoint; a sealed portion goes into D_final_test.

Speaker-attribution evaluation requirements. Contextual evaluation sets (pairs with two or more speakers, D_overlap) must let Phase 2 answer: who said the target; whether attacker and victim speech were distinguished; whether turn order was preserved; whether overlapping speakers were handled. Required data: target_speaker_id, role and turn timestamps for every utterance (section 15 schema). Evaluation concepts, no results yet:

Metric	Question it answers
Target-speaker identification accuracy	Was the target attributed to the right speaker?
Speaker attribution accuracy (per utterance)	Were context utterances attributed correctly?
Role-conditioned recall	DANGER recall when the context speaker is an attacker vs a friend vs none
Turn-order accuracy	Was attacker $\rightarrow$ victim distinguished from victim $\rightarrow$ attacker?
Overlap-specific recall	DANGER recall on targets inside an overlap span, by relative level (0/-6/-12 dB)

Role-conditioned recall is the direct test of the v2 failure (Phase 0 §11).

19. Data leakage prevention

Each check runs before a split is frozen and again when new data is added.

- ☐ Speaker leakage: no speaker ID (or voice-embedding near-duplicate across IDs) appears in two splits
- ☐ Recording leakage: one recording session and its augmented copies (6 conditions of one clip) stay in one split; count them as one item, not six
- ☐ Transcript duplication: exact and normalised transcript matches across splits flagged
- ☐ Template leakage: synthetic templates and scene scripts are split, not only their instances
- ☐ Near-duplicate audio: fingerprint match across splits and across public sources
- ☐ Same YouTube/media source in two splits: blocked by source ID
- ☐ Same scenario in train and test: scenario IDs split for D_hard_negative and D_final_test
- ☐ Synthetic contamination: no generator saw test items; generator outputs never in D_final_test
- ☐ Public-set overlap: IndicVoices-R and IndicVoices share audio; Vaani benchmark derives from Vaani — test portions of either family stay out of all training
- ☐ Phase 0 evaluation sets absent from every training manifest
- ☐ Pass rule for any sealed run written before the run (Phase 0 §22)

20. Data governance / licensing

This section flags items for legal review; it does not give legal conclusions. Items marked **LEGAL REVIEW** must be cleared before download for training or before collection starts.

Source / collection	Consent	PII / speaker identity	Raw audio storable?	Transcripts storable?	Commercial / redistribution	Flag
IndicVoices, Vaani, Vaani Noise	Collected by providers	Voices; district metadata	Per CC BY 4.0 + gate terms	Yes	Attribution; gate terms to read	LEGAL REVIEW of gate terms
Kathbath	Provider	Voices	Yes (CC0 packaging)	Yes	Underlying text from web crawl, per its card	Low

Source / collection	Consent	PII / speaker identity	Raw audio storable?	Transcripts storable?	Commercial / redistribution	Flag
MUCS Hi-En/Bn-En	Provider	Voices	Yes	Yes	CC BY-SA: share-alike on derived data	LEGAL REVIEW (share-alike reach)
Gram Vaani	Provider	Callers' voices	Academic use only	Academic	Commercial needs written permission	Request permission or eval-only
DISPLACE	Participants	Voices	Challenge terms only	Challenge terms	No sharing with non-participants	Not for product use
AMI	Provider	Voices	Yes	Yes	CC BY 4.0 (official site)	Keep official licence file
FSD50K	Freesound uploaders	Rarely voices	Per clip	—	Remove CC-BY-NC and Sampling+ clips for training; keep attributions	Low
IUEC (IIIT-D)	Volunteers	17 volunteers' daily life audio	Per EULA	—	Per EULA	LEGAL REVIEW
911 Recordings	Not given by callers	People in crisis; addresses possible	Eval only; restricted storage	Only derived labels	Licence unclear	LEGAL + ethics review
Custom collection	Written informed consent per speaker; separate consent for release, for acting violent scenes, and for voice reuse; withdrawal right	Speaker ID pseudonymous; demographic metadata stored apart from audio	Encrypted at rest (as Phase 0 does for journeys)	Yes, pseudonymous	Owned by project if contracts say so	LEGAL REVIEW of consent forms under India's DPDP Act

Source / collection	Consent	PII / speaker identity	Raw audio storable?	Transcripts storable?	Commercial / redistribution	Flag
User audio from production	Not collected for training	—	Not used	Not used	—	Out of scope without a new consent flow

Additional rules: adults only; actors are told scenes may be distressing and can stop at any time; no real victims are recruited or asked to recount real events; a deletion request removes the speaker's audio, transcripts and derived rows (including D_asr_errors) from every split, and the manifest records it.

Licence decision for sources Phase 2 may use. Based only on the evidence already cited in sections 3–11; nothing new was searched. A Hugging Face listing is not treated as proof of commercial rights.

Source	Status	Basis
Kathbath	Verified usable	CC0 packaging stated on the dataset card; underlying text from a web crawl
IndicVoices, Vaani, Vaani Noise Event	Usable only with conditions	CC BY 4.0 on card; attribution; gate conditions not yet read (legal review)
Lahaja	Usable only with conditions	CC BY 4.0 on card; attribution
AMI	Usable only with conditions	CC BY 4.0 on official site; OpenSLR mirror lists an older NC licence — use official source
BERSt	Usable only with conditions	CC BY 4.0 on card; attribution
MUCS 2021 sub-task 2	Usable only with conditions	CC BY-SA 4.0; share-alike reach on derived data under legal review

Source	Status	Basis
FSD50K	Usable only with conditions	Only CC0/CC-BY clips for training, per-clip licence files; NC and Sampling+ clips evaluation only
Gram Vaani	Requires permission	Academic use only; commercial needs written permission
IUEC (IIIT-Delhi)	Requires permission	EULA; terms not yet read
DISPLACE	Requires permission	Challenge-only terms; not for product use
Svarah, NOTSOFAR-1, 911 Recordings, CANNOT, HiACC	Evaluation/reference only	Licence unresolved from existing evidence; 911 also needs privacy review
Nonspeech7k	Evaluation/reference only	CC BY-NC-SA 4.0
SPRING-INX, MUCS sub-task 1, ASVP-ESD, VIVAE, IITKGP corpora	Not suitable until licence is resolved	Licence not established at source
MELD, AudioSet audio, WHAM! noise (in training mixes), Korean Emergency Audio	Not suitable	Copyrighted media, non-distributed audio, NC, or restricted access

21. P0 / P1 / P2 / P3 data requirements

P0 items block meaningful Phase 2 experiments: without them Phase 2 could only re-measure false alarms on English, which Phase 0 already did.

Priority	Data source	Phase 0 problem addressed	Why this priority
P0	Collected SAFE/DANGER pairs (En, Hi, Te, Hinglish, Te-En)	F4, F5-F10, F15	No public equivalent; core problem

Priority	Data source	Phase 0 problem addressed	Why this priority
P0	Collected attacker/friend → victim scenes with speaker IDs	F9, F10	The v2 failure cannot be tested without it
P0	Collected overlap + pocket scenes (D_overlap)	F2, F12, F13	Main recall gap; only 1 real example today
P0	IndicVoices (Hindi, Telugu, English-mixed subsets)	F3, F4	Needed to measure Indic ASR at all
P0	Vaani + Vaani Noise Event	F3, F4, F11, F13	Real Indian speech in real Indian noise, commercial-OK
P0	MUCS 2021 Hi-En	F4	Only sizeable open code-switched speech
P0	Frozen D_final_test incl. existing sealed sets	F15	Must exist before any model selection
P1	FSD50K CC0/CC-BY subset	F11	Scream positives + hard negatives
P1	Collected live-vs-replayed screams	F11	Unresolved Phase 0 problem 4
P1	AMI	F10, F12	Licence-clean turn + overlap structure
P1	BERSt	F11, F13	Shouted/fear speech on phones, CC BY
P1	D_asr_errors	F3, F8	Realistic STT errors for text models
P1	Kathbath, Lahaja, Svarah (eval)	F4	Clean per-language references
P1	Custom UDK recordings	§8	Never evaluated; placeholder live
P1	911 caller-only subset (eval)	F15	Only real emergency speech
P2	Gram Vaani (eval unless permission)	F4, phone band	Real telephone Hindi

Priority	Data source	Phase 0 problem addressed	Why this priority
P2	NOTSOFAR-1 (eval, licence pending)	F12	Real device overlap
P2	IUEC (if EULA allows)	F11	Indian distress contexts
P2	Other Indian CS pairs (Bn/Ta/Kn-En)	F4	Coverage beyond Hi/Te
P3	DISPLACE (if access granted, research only)	F10, F12	Indian multi-party overlap, restricted
P3	IITKGP emotion corpora, Nonspeech7k, VIVAE, ASVP-ESD (eval only)	F11	Acted or non-commercial

P0 acceptance criteria. A P0 item is complete only when every minimum below is met and audited from the manifest; having some examples is not enough. Thresholds differ by dataset on purpose. Current coverage for all collected items is 0 (nothing collected in Phase 1).

P0 item	Minimum required coverage	Current coverage
Type A SAFE/DANGER pairs	Allocation table in §15 met per language × context family; ≥ 40 unique speakers per language (≥ 25 for Telugu-English); ≥ 60% women; each pair from one speaker; SAFE:DANGER exactly 1:1; ≥ 4 environments and ≥ 3 device placements per language; ≥ 30% two-speaker and ≥ 10% overlap DANGER sides; provenance labelled on 100%	0 pairs

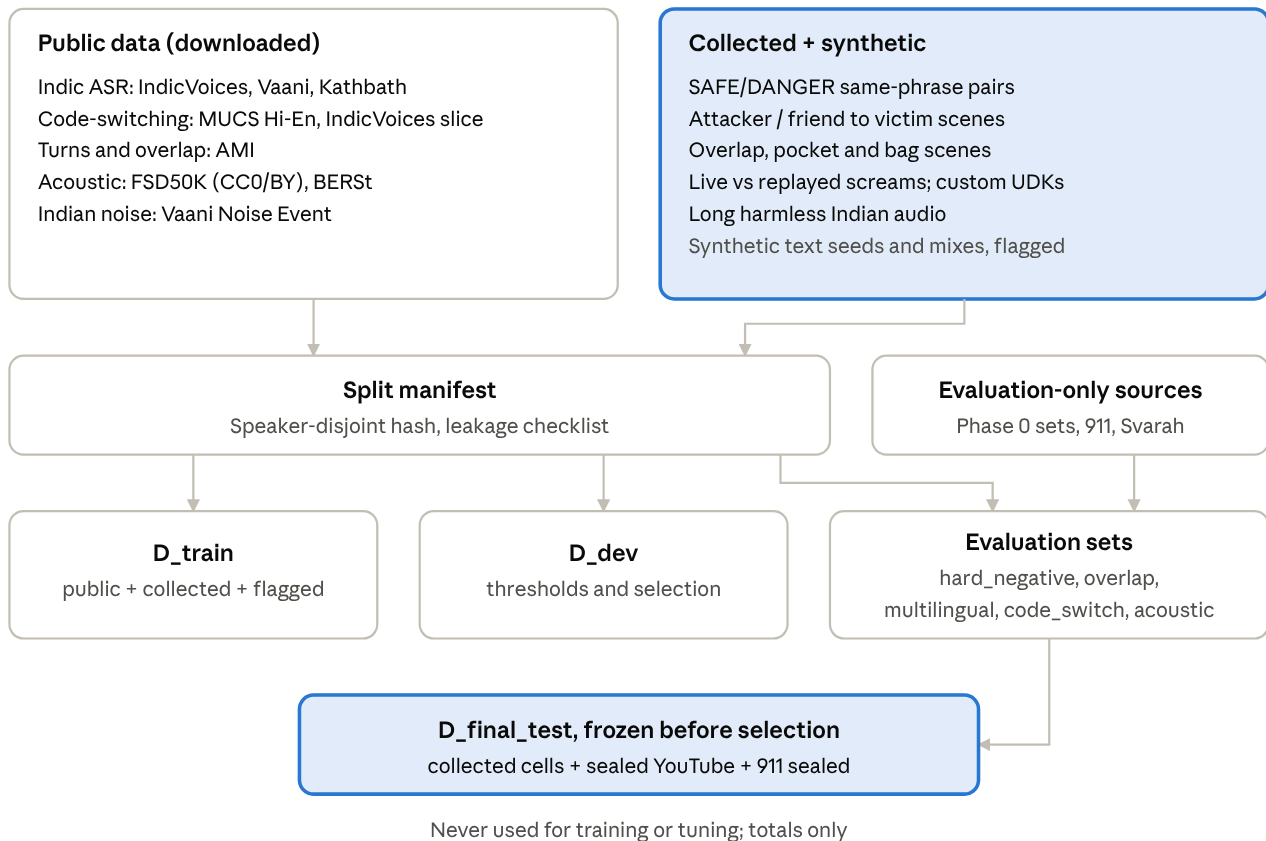
PO item	Minimum required coverage	Current coverage
Attacker/friend → victim scenes	≥ 150 attacker → victim and ≥ 150 matched friend → victim scenes with the same victim line; ≥ 30 victim speakers; victim and other speaker different people; roles and turn timestamps on 100%; both genders in attacker role	0 scenes
D_overlap (overlap + pocket/bag)	≥ 300 scenes; all four gender orders (M over F, F over M, same-gender pairs); each of 0/-6/-12 dB ≥ 25% of scenes; pocket, bag and hand each ≥ 20%; quiet, street, crowd each present; overlap timestamps on 100%	0 scenes
D_spontaneous_safe	As defined in section 18 (≥ 20 h, ≥ 30 participants, ≥ 6 environments)	0 h
Indic ASR evaluation (public)	Hindi and Telugu test sets from IndicVoices + Vaani with ≥ 50 speakers each; English-word script behaviour checked on a sample	Not yet downloaded
Code-switched evaluation	MUCS Hi-En test + blind; Te-En ≥ 2 h with ≥ 50 speakers (IndicVoices slice or collected); word-level language tags on 100% of eval items	Not yet downloaded; no Te-En tags exist
D_final_test	Frozen and hashed before any Phase 2 model selection; includes the Phase 0 sealed YouTube set and 911 sealed half unchanged, plus a stratified held-out share of every collected PO item above	Not frozen

P0 item	Minimum required coverage	Current coverage
Annotation quality (all collected P0)	Two annotators per item; disagreement adjudicated; agreement rate reported per field	Not started

22. Final data ecosystem

Public corpora feed ASR, code-switching, turn/overlap mechanics and acoustic hard negatives; collected and flagged-synthetic data carry the safety contexts. Everything passes one split manifest; evaluation-only sources skip training entirely.

Public data teaches mechanisms; collected pairs carry the safety problem



target data ecosystem · sources → manifest → splits → frozen test

The collected box is highlighted because it holds the only data that tests the project's central problem; the frozen test set is fixed before any Phase 2 model is chosen.

23. Phase 2 inputs

Phase 2 receives the sources below. Licence classes and limitations are as sourced in sections 3–11; every NOT VERIFIED licence must be confirmed before that source moves from evaluation to training.

Data source	Purpose	Subset	License	Usable split	Preprocessing required	Known limitations
IndicVoices	Indic ASR train/eval; CS slice	Hindi, Telugu, English-mixed; CoSTA-style high-CMI slice	CC BY 4.0	Train + test by source speakers	Check English-word script; add word-level tags; resample to 16 kHz	Short utterances; no safety content
Vaani (transcribed part)	Spontaneous Indic ASR	Hindi, Telugu districts	CC BY 4.0, gated	Speaker-disjoint split we define	Filter by transcription availability; language ID per clip	~10% transcribed
Vaani Noise Event	Noise robustness; acoustic hard negatives	All 7 categories	CC BY 4.0, gated	Released portion (authors hold back 10 h eval)	Use timestamps to cut events; SNR estimate	Background events, not distress
MUCS 2021 sub-task 2	Hi-En CS ASR	Hi-En train/test/blind	CC BY-SA 4.0	Given splits	Word-level tags from script; normalise	Lecture domain
Kathbath	Clean read references	Hi, Te + others	CC0	Given splits	m4a → wav	Read speech
Lahaja, Svarah	Accent evaluation	All	CC BY 4.0 / NOT VERIFIED	Eval only	None	Small
FSD50K	Scream/cry + hard negatives	CC0 + CC-BY clips of target classes	CC BY (dataset) + per clip	Given dev/eval	Licence filter by clip JSON	Not Indian; weak labels
AMI	Turns, overlap	Headset + far-field	CC BY 4.0	Official splits	Per-speaker alignment	Office English
BERSt	Shouted/fear speech, phone positions	All	CC BY 4.0	Speaker-disjoint	None	Acted, nonsense phrases

Data source	Purpose	Subset	License	Usable split	Preprocessing required	Known limitations
911 caller-only subset	Real emergency recall	Dev half only until sealed run	NOT VERIFIED	Eval only	Remove dispatcher segments	US English
Collected corpus	Pairs, scenes, overlap, pocket, custom UDKs, live/replayed	All cells of §15	Own (per consent)	Speaker-disjoint, frozen	Double annotation	Acted; pilot first
D_asr_errors	STT error modelling	Built from above	Inherits each source	Mirrors source split	Frozen-engine inference only	Depends on source coverage

Data we will download: IndicVoices, Vaani (transcribed part), Vaani Noise Event, MUCS 2021 sub-task 2, Kathbath, Lahaja, FSD50K, AMI, BERSt.

Data we will collect: SAFE/DANGER target-phrases; attacker/friend/narrator → victim scenes; overlap and pocket/bag scenes; live vs replayed screams; frightened speech; custom UDK phrases; long continuous harmless Indian audio from consenting participants.

Data we will synthesize: text seeds for pairs and code-switched scripts (reviewed by native speakers), negation pairs, noise and overlap mixes from licence-clean audio, simulated ASR errors checked against D_asr_errors.

Data we will use only for evaluation: Phase 0 prepped_data, real_recordings, dev and sealed YouTube, Telugu own recordings, 911 (both halves; sealed stays sealed), Svarah, Gram Vaani (unless permission), NOTSOFAR-1 (until licence checked), IUEC (per EULA), IITKGP emotion corpora, Nonspeech7k, VIVAE, ASVP-ESD.

Data we will not use: AudioSet audio re-downloaded from YouTube, MELD audio, DISPLACE for anything product-facing, WHAM! noise in training mixes, Korean Emergency Audio (restricted), LLM-generated context rows from intent v2 as-is.

Decision gate. Based on the available evidence, these are the data resources required before Phase 2 can begin.

Phase 1 Closure

Completed

- Data requirements mapped to every Phase 0 failure (§2, §13), with P0–P3 priorities (§21).
- Public sources identified, with what each can and cannot provide (§3–11, §14).
- Collection design: speakers, languages, devices, scenario matrix, proposed 3,000-pair allocation, provenance labels, three separate problem types (§15).
- Representation of pairs, speakers, roles, turns, overlap, code-switching, context_before/context_after and timestamps (§5, §8, §15).
- Synthetic data rules and ASR-error resource (§16–17).
- Splits, D_spontaneous_safe, speaker-attribution metrics, leakage checklist, frozen-test principle (§18–19).
- Licence decision per usable source and governance flags (§20); P0 acceptance criteria with current coverage (§21).

Remaining before Phase 2

These are prerequisites to execute, not open specification questions:

- ☐ Legal review: gate terms (IndicVoices, Vaani), MUCS share-alike reach, consent forms under the DPDP Act, 911 privacy
- ☐ Metadata check: whether IndicVoices and Vaani keep English words in Latin script
- ☐ 20-speaker collection pilot, then confirm or adjust the §15 allocation
- ☐ Collect P0 items to the §21 acceptance criteria
- ☐ Freeze and hash D_final_test before any Phase 2 model selection

Phase 2 work on public data (Indic/code-switched ASR evaluation, D_asr_errors) can start once the downloads and legal review are done; Phase 2 experiments on context and speaker attribution cannot start until the collected P0 items meet their acceptance criteria.

Phase 2 inputs

The Phase 2 inputs table and download/collect/synthesize/eval-only/not-used lists (§23), the §15 annotation schema and pair allocation, the §18 split definitions and metrics, the §20 licence decisions and the §21 acceptance criteria.

Phase 1 status

READY FOR PHASE 2 — all closure requirements are specified; data collection and the prerequisites above are Phase 2 entry work, not Phase 1 gaps.